



ANNUAL DAVAO CITY AIRSHED STATUS REPORT **2023**

ENVIRONMENTAL
MANAGEMENT BUREAU XI

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Executive Summary

This report provides a general overview of the Davao City Airshed in terms of location, topography, population, land use, air quality monitoring (CY 2021-2023), and emission inventory (CY 2022). This report was prepared to reflect the air quality status of the Davao City Airshed for CY 2023.

Between 2015 and 2020, the population of Davao City increased from 1,629,843 to 1,825,450 in total with an annual population growth rate of 2.27% (PSA, 2020).

Based on the 3-year air quality monitoring conducted by the Environmental Management Bureau XI within the Davao City Airshed (DCA) or Davao City Air Quality Monitoring Network (DCAQMN), most of the concentration of parameters for 24-hour monitoring is within the short-term guideline values.

The Davao City Emission Inventory for CY 2022 shows that most of the pollutants are coming from mobile sources. These pollutants include Particulate Matter (PM), Carbon Monoxide (CO), Nitrogen Oxides (NO_x), and Volatile Organic Compounds (VOCs). When compared to other sources of pollutants, mobile sources contributed 56%, 96%, 68%, and 98% of the total concentration for PM, CO, NO_x, and VOCs, respectively. On the other hand, the highest Sulfur Oxide (SO_x) emissions are from stationary sources using fossil fuels which contribute 98% of the total concentration.

In CY 2023, the Davao City Airshed Governing Board has come up with strategies to further strengthen the monitoring and management of the air quality within the airshed. Eight (8) board resolution were passed by the board, which are the following: (1) Confirmation of the Representatives of Active Sectoral Groups through Securities and Exchange Commission (SEC) Certification and Appointment of Permanent and Alternate Representatives for the Davao City Airshed Governing Board, (2) To Include Davao Inventors Association (DIA) as a Regular Member of the Davao City Airshed Governing Board, Represented by Mr. Virgilio M. Sangutan, (3) Proposal that Non-Active Members be Removed from the Davao City Airshed Governing Board List of Members, (4) Creation of Technical Working Group for the Davao City Airshed 5-Year Action Plan CY 2025 to CY 2029, (5) Special Meeting of the Technical Working Group to Draft the Davao City Airshed 5-Year Action Plan CY 2025 to CY 2029, (6) The Inclusion of Women's Group or Environmental Group with GAD-Related Program in the Governing Board be Deferred, (7) Requiring the Sectoral Members to Submit General Information Sheet (GIS) Duly Received by Securities and Exchange Commission (SEC), and (8) Terms of Membership of Active will be Renewed while Membership of inactive Sectoral Members will be Subject for Evaluation and Communication for their Interest to Sit as Governing Board Members

As to the monitoring of vehicular emissions, the Davao City Airshed Governing Board through the Anti-Smoke Belching Unit (ASBU) of the Davao City Environment and Natural Resources Office (CENRO), has conducted monthly roadside inspections. A total of 526 citation tickets were issued to drivers/operators showing non-compliance with emission standards.

Based on the result of monitoring of all criteria pollutants for CY 2023, the present general ambient air quality of Davao City Airshed still ranges from **Good to Fair**.

INTRODUCTION

The Republic Act 8749, otherwise known as the Philippine Clean Air Act (CAA) of 1999, requires the Department of Environment and Natural Resources (DENR) through the Environmental Management Bureau (EMB) to designate an airshed that will be managed by multi-sectoral governing board which is tasked to formulate policies and standards as well as actions plans to manage air quality situations within the area.

An airshed refers to areas with common weather or meteorological conditions and sources of air pollution that affect the interchange and diffusion of pollution in the surrounding atmosphere.

On February 12, 2003, through Memorandum Circular No. 2003-02, Davao City was designated as an Airshed of Region XI, Pursuant to Rule XV, Section 2 of the CAA. Davao City was established as an Airshed of Region XI considering that the city is a highly urbanized area and the economic hub of the region with a vast number of industries with potential sources of air pollution and with the highest number of population and dwellers.

In CY 2023, the DENR, through EMB XI, has continuously monitored the ambient air quality of the airshed through the Davao City Air Quality Monitoring Network (DCAQMN) established in 2013 which is composed of four (4) manual monitoring stations and two (2) real-time air quality monitoring stations. All the data gathered for all the monitored criteria pollutants comply with at least 75% data capture rate requirement from all the monitoring stations, representing the 2023 Annual Air Quality.

In the same year, the Davao City Airshed Governing Board formulated an Action Plan for the activities to be undertaken to operationalize the airshed. Activities include a series of meetings to come up with strategies for maintaining the air quality of the airshed; strengthening the monitoring of vehicular emissions; and continuous Information, Education, and Communication (IEC) campaign.

CHARACTERISTICS OF DAVAO CITY AIRSHED

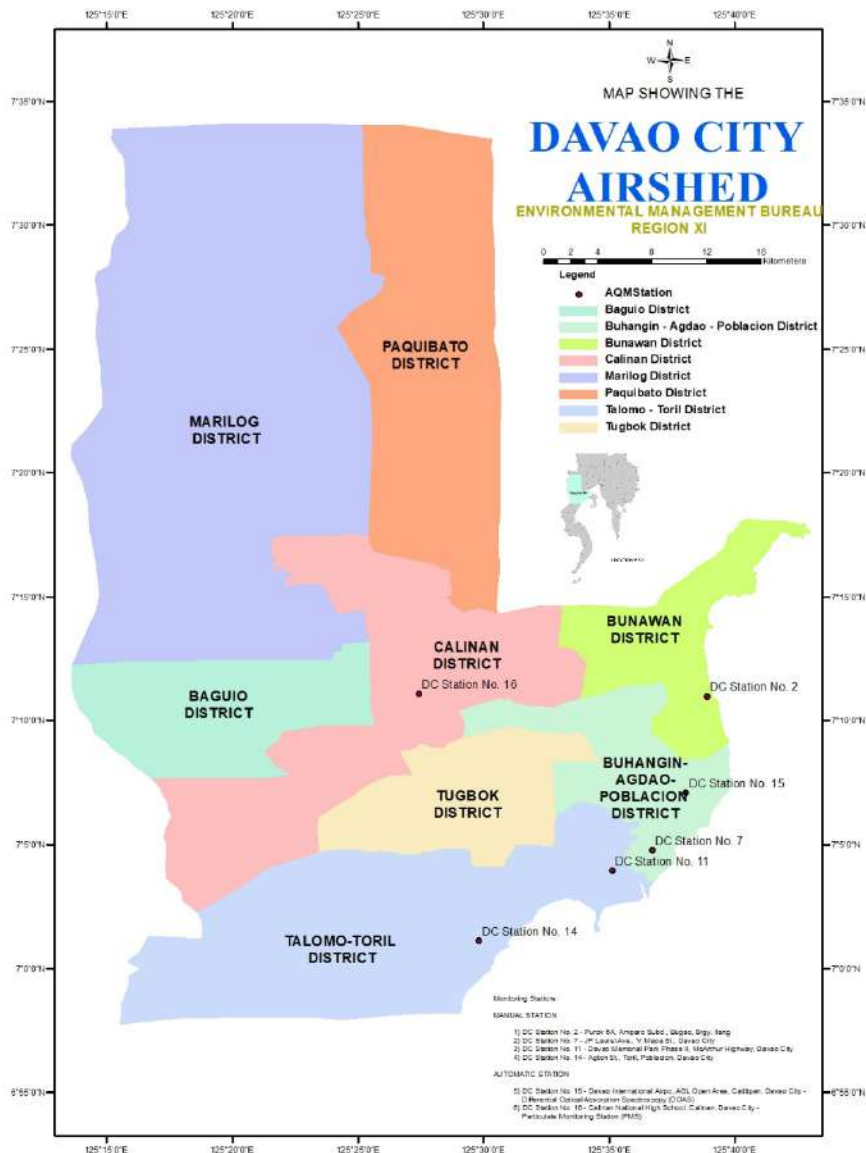
Area and Location

Davao City has a total area of 244,000 hectares and is tagged as the largest city in the Philippines in terms of land area. The city is divided into 3 congressional districts (Districts 1, 2 and 3) which are further divided into 11 administrative districts with 182 barangays. It is geographically located in the southeastern part of Mindanao, on the northwestern part of Davao Gulf. The city was developed as the regional trade center in southern Mindanao due to its strategic location.

Table 1. Davao City Districts

Congressional District	Administrative District
District 1	Poblacion, Talomo
District 2	Agdao, Buhangin, Bunawan, Paquibato
District 3	Baguio, Calinan, Marilog, Toril, Tugbok

Figure 1 shows the whole Davao City Airshed Map, which is geographically located in the grid squares of 6° 58' to 7° 34' N latitude and 125° 14' to 125° 40' E longitude.



Topography and Climate

The southeast area of the city is plain and slightly hilly along the entire coast and uplands north and westward to Calinan, with slopes generally below four to five degrees. The plains and valleys merged gradually into the uplands and in turn into the mountains. Although the eastern part of the city is a broad lowland belt, its surface is interspersed by low hills and knobs. (source: *davaocity.gov.ph*, 2020)

Davao City has a tropical rainforest climate based on *Koppen Climate Classification*.

Population

Based on the 2020 Population Census, Davao City has a population of 1,825,450. The said population is higher by 192,459 when compared to the 2015 Population Census which is 1,632,991. The annual population growth rate of Davao City based on the 2015-2020 reference period is 2.27%. (source: *PSA*, 2020)

Table 2. Davao City Population

	Population (in thousands)		Annual Population Growth Rate
	2015	2020	
Davao City	1,633	1,825	2.27%

Land Use

Based on the Comprehensive Land Use Plan of Davao City (2013-2022), land use in the airshed is predominantly Agricultural Use with a total of 111,466 hectares or 45.66% of the total. Major industrial activities are located in the northern part of Davao City in Bunawan District while commercial activities are located in the center of the city mostly in Agdao and Poblacion Districts.

AIR QUALITY IN DAVAO CITY

Ambient Air Quality Monitoring

The established Davao City Air Quality Monitoring Network (DCAQMN) in the Airshed is an area quality monitoring network (AQMN) of urban scale representation composed of selected sampling sites from the three (3) AQMNs of neighborhood scale representation which are: 1) Buhangin-Agdao-Poblacion District AQMN; 2) Bunawan District AQMN; and 3) Talomo-Toril District AQMN.

In 2014, additional monitoring stations were established in the Calinan and Buhagin Districts which represent an urban scale. The purpose of the establishment of the AQMN is to make a continuous air quality trend within the airshed which is useful for researchers, decision-makers, and lawmakers.

In 2023, six (6) monitoring stations (4 - using manual equipment and 2 - using real-time equipment) were maintained and operated by EMB XI. These stations were able to monitor Particulate Matter 10 and 2.5. One of the two (2) real-time monitoring stations also determined the level of pollutant Gases (SO₂, NO₂, O₃, CO, and BTX) concentration in the ambient air.

Table 3. Davao City Air Quality Monitoring Stations

Sampling Stations	Type of Monitoring	District	Sampling Type	Sampling Frequency	Criteria Pollutants	Spatial Scale
DC Station No. 2	Long-term trending	Bunawan	Manual	Every 6 days	PM-10	Neighborhood
DC Station No. 17	Roadside	Buhangin-Agdao-Poblacion	Manual	Every 6 days	PM-10	Neighborhood
DC Station No. 11	Long-term trending	Talomo-Toril	Manual	Every 6 days	PM-10	Neighborhood
DC Station No. 14	Roadside	Talomo-Toril	Manual	Every 6 days	PM-10	Neighborhood
DC Station No. 15	General Ambient	Buhangin-Agdao-Poblacion	Automatic	Continuous	PM-2.5, PM-10, SO ₂ , NO ₂ , O ₃ , CO, BTX	Neighborhood
DC Station No. 16	General Ambient	Calinan	Automatic	Continuous	PM-2.5 and PM-10	Neighborhood

- Neighborhood Scale - defines concentrations within some extended area of the city that has relatively uniform land use with dimensions in the 0.5 – 4.0 kilometers range.
- Urban Scale - defines concentrations within an area of city-like dimensions on the order of 4.0-50.0 kilometers.

Air Quality Monitoring Station

Manual Stations

DC Station No. 2

DC Station No. 2 is located at Purok 5A, Pagasa Village, Bugac, Ilang, Davao City. It is approximately 70 meters from Amparo Road and 300 meters from the Davao-Agusan National Highway. At the south of the site are major industries and residential communities along the Panacan and Ilang areas. On the eastern side are the Davao Agusan National Highway, dense residential areas, and private wharves. On the western side of the site are also residential areas while on the north side are residential subdivisions, commercial establishments, and industries. Air pollutants coming from the emissions of these major industries along Panacan and Ilang Areas together with the emissions of motor vehicles plying the National Highway, and water vessels that dock along the different wharfs are expected to be measured in this station. A Memorandum of Agreement (MOA) was signed by EMB XI on September 27, 2023, with Holcim Philippines – Davao Plant to approve the establishment of an unmanned ambient air quality monitoring station within the Barangay Ilang, Holcim Relocation Area, PAGASA Phase II.

DC Station No. 17

DC Station No. 17 is located within the open botanical garden of Marfori Heights Subdivision, Brgy Maa, Poblacion District, Davao City replacing DC Station No. 07 on JP Laurel Ave., V. Mapa Street, Davao City. This station represents the emissions of road vehicles, residences, and commercial establishments. Approximately 40 to 600 meters radius are commercial establishments, schools, residential subdivisions, and cemeteries. It is approximately 30 meters from Ruby Street, Topaz Street, Diamond Street, and Garnet Street. The site is categorized as roadside and will be monitored for PM₁₀.

DC Station No. 11

DC Station No. 11 is located at Davao Memorial Park, Phase II, McArthur Highway, Davao City. South of the site, approximately 34 meters, is the Morales Subdivision. Northeast of the station is a cluster of industries, bus garages (along Ma-a Road), residential areas, and construction activities of First Lake Memorial Park. At the southwest of the site, approximately 300 meters and 180 meters respectively are a hill and residential sources. South of the site approximately 644 meters is the McArthur Highway. This site was selected to monitor emissions of sources along Ma-a Road, McArthur Highway, and emissions of surrounding residential areas. Also, the site will monitor for a possible build-up of pollutants at the foot of the hill when the prevailing wind direction is towards the southwest.

DC Station No. 14

DC Station no. 14 is located at Open Park Ground, Agton St., Toril Poblacion, Davao City. To the northwest of the site is the Public Market of Toril. On the south and southwest side are commercial establishments. On the North side is the Davao-Cotabato

National Highway and on the East side are commercial establishments along Agton Street. The site is representative of road vehicle, residential, and commercial emissions.

Real-Time Monitoring Stations

DC Station No. 15

DC Station No. 15 is located at Davao International Airport, AGL Open Area, Catitipan, Buhangin District, Davao City. It is more than 500 meters from the Maharlika National Highway. The station is a representative of industrial, road vehicle, aircraft, watercraft, residential and commercial emissions.

DC Station No. 16

DC Station No. 16 is located in the Open Area near the Oval playing field of Calinan National High School, Calinan, Davao City. It is approximately 500 meters from the Davao-Bukidnon National Highway. The station is a representative of road vehicle, residential, and commercial emissions.

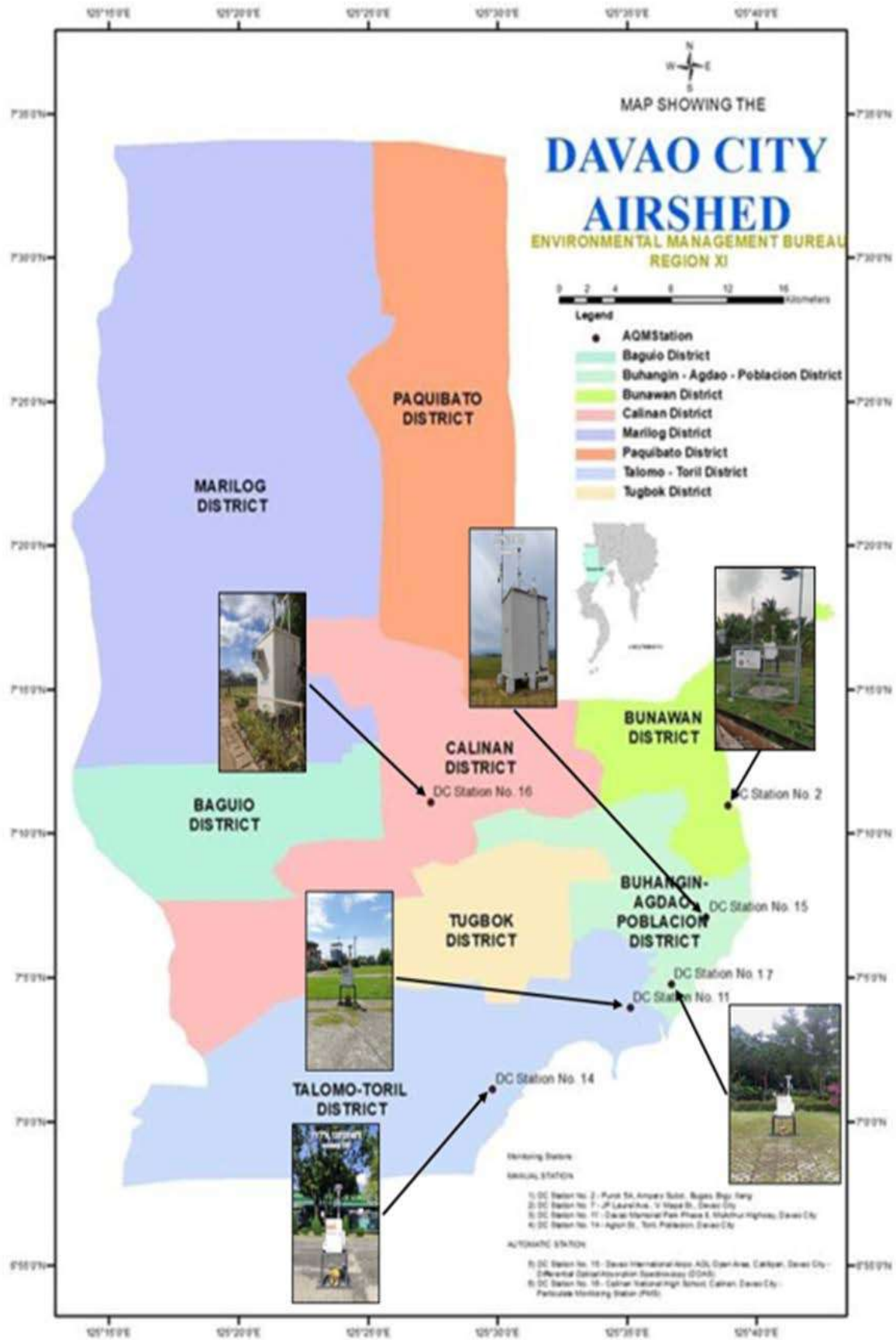


Figure 2. Davao City Airshed Map with Air Quality Monitoring Stations

AIR POLLUTANTS MONITORED (AMBIENT)

Suspended Particulate Matter (PM₁₀) and (PM_{2.5})

Particulate matter can be a significant air pollutant that is associated with a variety of health and environmental effects. Typical air pollution sources in the network are (1) fine particulates emitted as a result of fuel combustion from road vehicles, power generation, industrial processes, and commercial and domestic activities, (2) fine particulates arising from a wide range of sources, including re-suspended dust by road vehicles, construction works, wind-blown dust and soils, sea salts and biological particles such as pollen.

The primary concerns are namely the PM-10 (Particulate Matter with a diameter of less than 10 micrometers but larger the 2.5 micrometers) and PM-2.5 (Particulate Matter with a diameter of less than 2.5 micrometers).

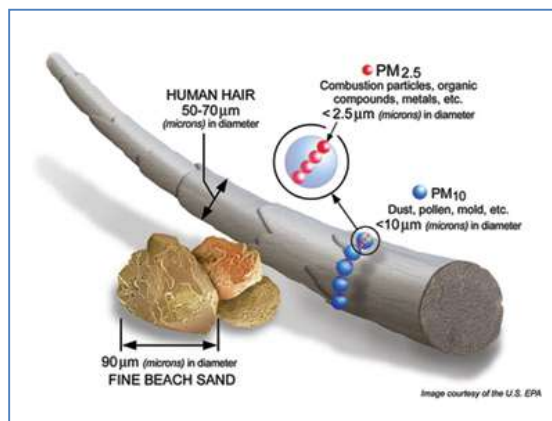
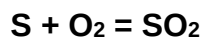


Figure 3. Particulate Matter Size

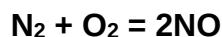
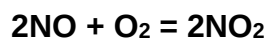
Sulfur Dioxide (SO₂)

Sulfur dioxide in the network comes mostly from the emissions of motor vehicles and the burning of fossil fuels from industries and standby generating sets from commercial establishments. Sulfur dioxide is the result of the combustion of sulfur-containing fuels: namely coal, bunker, and diesel.



Nitrogen Dioxide (NO₂)

Nitrogen Dioxide (NO₂) is one of the group of highly reactive gasses known as oxides of Nitrogen (NO_x.) It was selected as the indicator for the larger group of NO_x. Nitrogen Dioxide is a concern due to its potential to cause health effects, and the monoxide form, nitrogen oxide (NO), which is less toxic but may oxidize to NO₂ in the atmosphere. Nitrogen oxide is mainly produced from the combustion of nitrogen using air as an oxidant at elevated temperature and pressure. This then oxidizes to nitrogen dioxide at atmospheric conditions but is a very slow process.



Major sources of this pollutant come from motor vehicles and large combustion units from industrial processes. Since nitrogen oxides are formed from the combustion

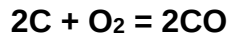
of nitrogen with air, almost any engine operating at elevated temperature and pressure produces these oxides.

Ground Level Ozone (O₃)

This ozone is formed at ground level through a chemical reaction of various air pollutants with sunlight. Ozone is rarely emitted directly into the air. The pollutants that contribute to ozone formation are oxides of nitrogen (NO_x) and volatile organic compounds (VOCs). Major sources of these pollutants in the network are vehicle and engine exhaust, emissions from industrial facilities, combustion from significant electric generating sets, gasoline vapors, and chemical solvents.

Carbon Monoxide (CO)

Carbon Monoxide is a colorless, odorless gas that is very toxic to humans even at low concentrations. It is a product of the incomplete combustion of carbon, which occurs when air is not sufficient to facilitate the complete combustion of carbon from carbon-containing fuels. Primary sources are from motor vehicle exhausts, which contribute the most CO emissions nationwide.



CO, in the presence of solar radiation, reacts with other chemical compounds to form ground-level ozone.

AMBIENT AIR MONITORING RESULT

Manual Monitoring

Particulate Matter 10 (PM_{10})

Below is the Time Plot History Graph from CY 2021 to CY 2023 showing the 24-hr PM_{10} concentrations at Station Nos. 2,7,11 and 14 are still within the short-term guideline value of 150 ug/Ncm.

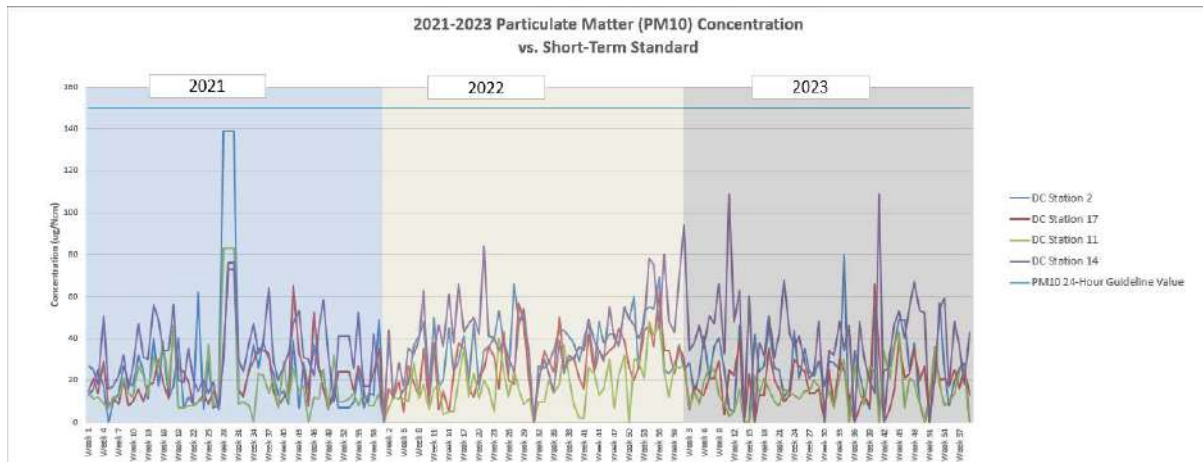


Figure 4. Time Plot History Graph from CY 2021 to CY 2023 of PM_{10}

Table 4. Summary Result of 24-Hour Concentration of PM_{10} from CY 2021 to CY 2023

Davao City Monitoring Station	24-Hour Concentration (ug/Ncm)						Short-Term Guideline Value (ug/Ncm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 2	6.00	139.00	11.00	69.00	5.00	80.00	150
Station 17	7.00	73.00	5.00	62.00	4.00	66.00	
Station 11	7.00	83.00	2.00	48.00	3.00	43.00	
Station 14	6.00	76.00	11.00	84.00	14.00	109.00	

As shown in Table 5, the annual geometric mean for PM_{10} for CY 2021 to CY 2023 shows that the concentration from all monitoring stations recorded is within the long-term guideline value.

Table 5. Summary Result of Annual Geometric Mean of PM_{10} from CY 2021 to CY 2023 (Manual Stations)

Davao City Monitoring Station	Annual Geometric Mean Concentration (ug/Ncm)			Long-Term Guideline Value (ug/Ncm)
	2021	2022	2023	
Station 2	18.73	33.51	26.00	60
Station 17	18.53	24.59	20.00	
Station 11	15.02	14.92	16.00	
Station 14	28.99	38.03	47.00	

Real-Time Monitoring

Particulate Matter 10 (PM_{10})

Figure 5 shows the Time Plot History Graph from CY 2021 to CY 2023 showing that the 24-hr PM_{10} concentrations at Station Nos. 15 and 16 are still within the short-term guideline value of 150 ug/Ncm.

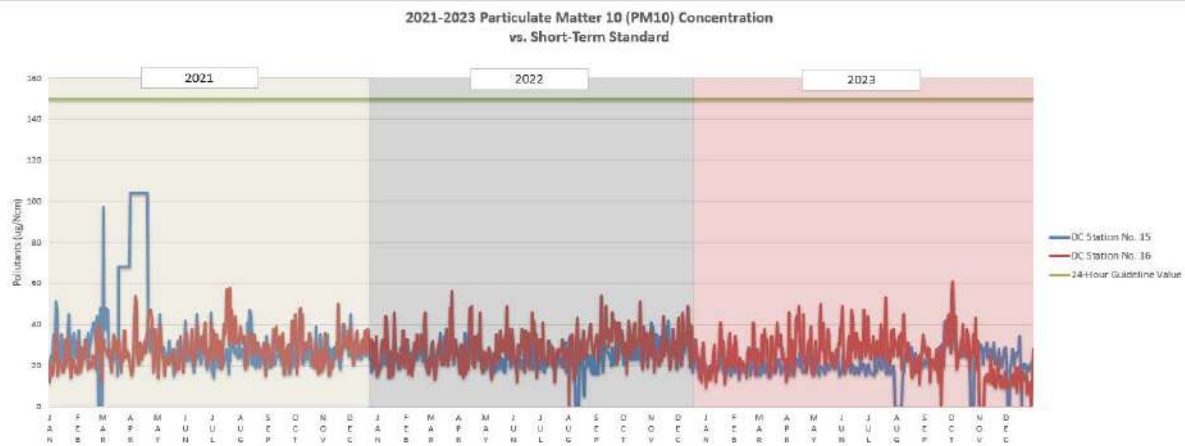


Figure 5. Time Plot History Graph from CY 2021 to CY 2023 of PM_{10} (Real-time Monitoring Station)

Table 6. Summary Result of 24-Hour Concentration of PM_{10} from CY 2021 to CY 2023

Davao City Monitoring Station	24-Hour Concentration (ug/Ncm)						Short-Term Guideline Value (ug/Ncm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 15	14.00	104.00	5.00	43.00	13.00	45.00	150
Station 16	12.00	58.00	12.00	56.00	6.00	61.00	

In CY 2021 - CY 2023, the targeted monitoring data for Station 16 and Station 15 are above the minimum 75% requirement as outlined in Memorandum Circular 2005-13 Section 4(3). This makes them representative data for the stated years.

Table 7. Summary Result of Annual Geometric Mean of PM_{10} from CY 2021 to CY 2023 (Real-Time Monitoring Stations)

Davao City Monitoring Station	Annual Geometric Mean Concentration (ug/Ncm)			Long-Term Guideline Value (ug/Ncm)
	2021	2022	2023	
Station 15	29.69	24.29	21.00	60
Station 16	26.89	24.07	24.00	

Particulate Matter 2.5 ($PM_{2.5}$)

Below is the Time Plot History Graph from CY 2021 to CY 2023 showing that most of the 24-hr $PM_{2.5}$ concentrations at Station Nos. 15 and 16 are within the short-term guideline value of 50 ug/Ncm and 35 ug/Ncm.

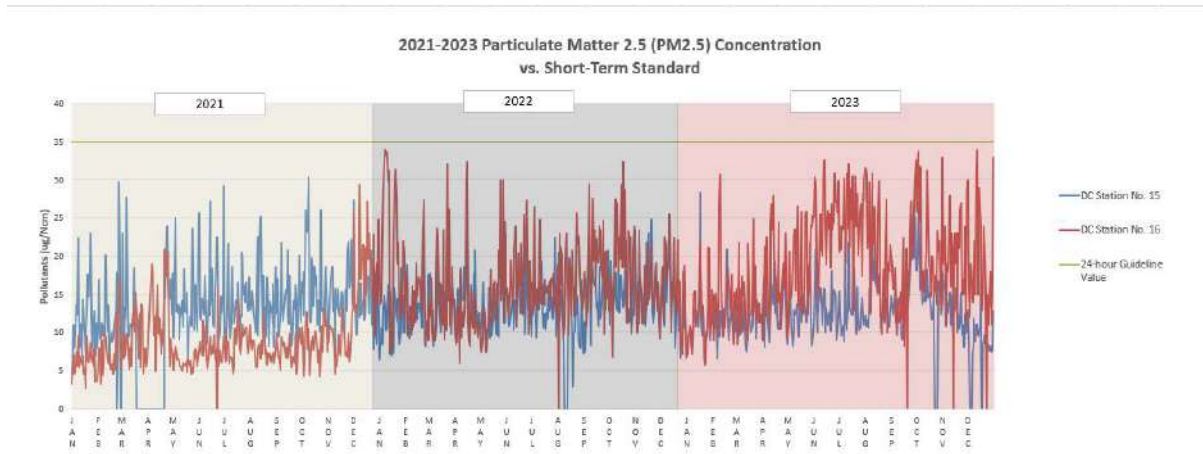


Figure 6. Time Plot History Graph from CY 2021 to CY 2023 of PM_{2.5} (Real-time Monitoring Station)

Table 8 shows the minimum and maximum concentration of both monitoring stations from CY 2021 to CY 2023. As of now, there is no air quality index for PM_{2.5}.

Table 8. Summary Result of 24-Hour Concentration of PM_{2.5} from CY 2021 to CY 2023

Davao City Monitoring Station	24-Hour Concentration (ug/Ncm)						Short-Term Guideline Value (ug/Ncm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 15	6.69	35.48	2.90	5.90	6.40	28.30	35
Station 16	2.36	31.19	25.20	34.00	5.80	34.00	

Table 9. Summary Result of Annual Geometric Mean of PM 2.5 from CY 2021 to CY 2023 (Real-Time Monitoring Stations)

Davao City Monitoring Station	Annual Geometric Mean Concentration (ug/Ncm)			Long-Term Guideline Value (ug/Ncm)
	2021	2022	2023	
Station 15	13.33	13.37	12.70	25
Station 16	8.37	15.66	18.20	

Sulfur Dioxide (SO₂)

Below is the Time Plot History Graph from CY 2021 to CY 2023 showing that the maximum values of SO₂ recorded at Station No. 15 for the 3 consecutive years are within the short-term guideline value of 180 ug/Ncm.

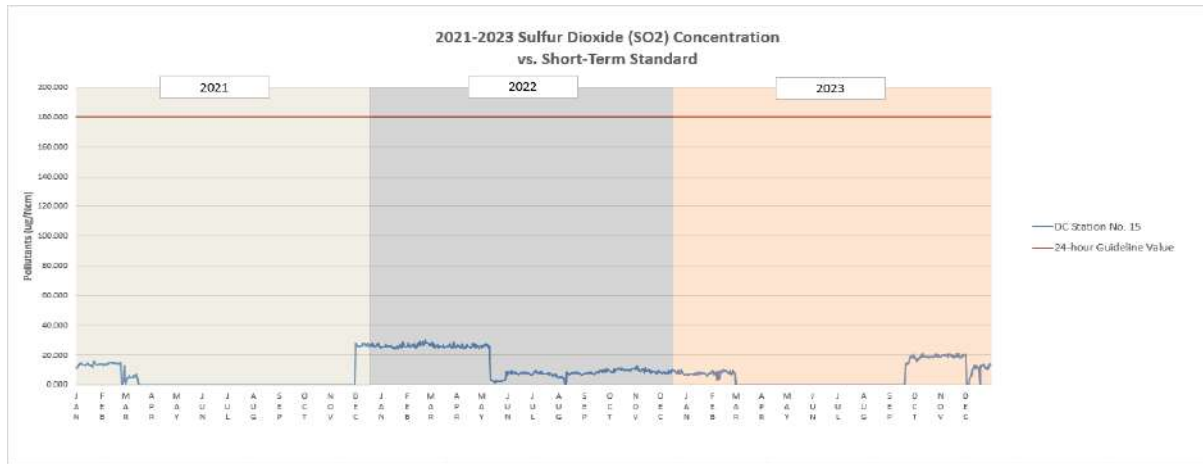


Figure 7. Time Plot History Graph from CY 2021 to CY 2023 of SO₂

Note:

- 1) Insufficient data for CY 2021 and CY 2023 due to defective/malfunctioned equipment

Table 10. Summary Result of 24-Hour Concentration of SO₂ from CY 2021 to CY 2023

Davao City Monitoring Station	24-Hour Concentration (ug/Ncm)						Short-Term Guideline Value (ug/Ncm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 15	1.89	28.20	1.71	30.04	2.89	21.35	180

The annual arithmetic mean for SO₂ for CY 2021 to CY 2023 shows that the concentration is within the long-term guideline value of 80 ug/Ncm.

Table 11. Summary Result of Annual Arithmetic Mean of SO₂ for CY 2021 to CY 2023

Davao City Monitoring Station	Annual Arithmetic Mean Concentration (ug/Ncm)			Long-Term Guideline Value (ug/Ncm)
	2021	2022	2023	
Station 15	13.93	11.60	6.83	80

Nitrogen Dioxide (NO₂)

Below is the Time Plot History Graph from CY 2021 to CY 2023 showing that the maximum values of NO₂ recorded at Station No. 15 for the 3 consecutive years are within the short-term guideline value of 150 ug/Ncm.

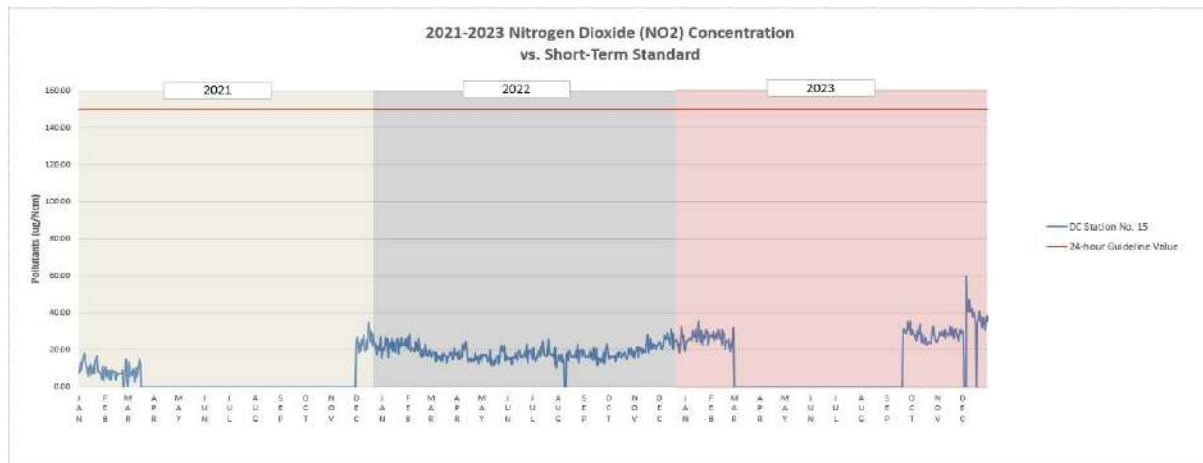


Figure 8. Time Plot History Graph from CY 2021 to CY 2023 of NO2

Note:

- 1) Insufficient data for CY 2021 and CY 2023 due to defective/malfunctioned equipment

Table 12. Summary Result of 24-Hour Concentration of NO2 from CY 2021 to CY 2023

Davao City Monitoring Station	24-Hour Concentration (ug/Ncm)						Short-Term Guideline Value (ug/Ncm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 15	1.56	35.017	10.22	32.34	18.7	59.86	150

Carbon Monoxide (CO)

The table shows that the 8-hour concentration of CO has a gradual increase from CY 2021 to CY 2023. A table for 1-hour concentration of CO for CY 2023 was also presented which has a short-term guideline value of 30ppm.

Table 13. Summary Result of 8-Hour Concentration of CO from CY 2021 to CY 2023

Davao City Monitoring Station	8-Hour Concentration (ppm)						Short-Term Guideline Value (ppm)
	2021		2022		2023		
	Min	Max	Min	Max	Min	Max	
Station 15	0.214	8.450	0.000	4.027	0.031	3.140	9

Table 14. Summary Result of 1-Hour Concentration of CO for CY 2023

Davao City Monitoring Station	1-Hour Concentration (ppm)		Short-Term Guideline Value (ppm)
	2023		
	Min	Max	
Station 15	0.010	3.670	30

Davao City Emission Inventory

Emission Inventory (EI) is a comprehensive listing of pollutants from all sources in a geographical area contributing to the atmosphere. The tables of EI are not only useful for regulatory agencies, but it is also necessary for planning and zoning sources. In addition, it can be used to identify the major contribution sources. Control strategies can be developed to a considerable reduction of pollution in the areas. Emission inventory is an essential tool to understand the air quality from local to regional, and to global scales. An emission inventory is comprised of a list of relevant air pollutants in a specific area, broken down by activity sector, for example, traffic, industry, and construction. The emission inventory is conducted every three years based on guidelines, thus, the latest emission inventory was that of 2022.

Data are collected during an emissions test to verify that the process is operating at the desired production level (USEPA, 2021). If there is no detailed data available, the activity data can be estimated from related available data using several assumptions (UNEP/MOE, 2001). The 2022 Davao City Emission Inventory is composed of several data summarized in the table below.

Emission Sources	Data Source
Stationary	Environmental Management Bureau XI
Area	
Aircrafts	Civil Aviation Authority of the Philippines
Fires (Automobile and Structural)	Bureau of Fire Protection XI
Gasoline Dispensing	Department of Energy
Mobile	Land Transportation Office XI

2022 Davao City Emission Inventory

As shown in the table below the highest pollutant emitted within Davao City Airshed in CY 2022 is Carbon Monoxide (CO) with an emission rate of 119,343.21 tons per year followed by Volatile Organic Compounds (VOCs) with an emission rate of 56,216.20 tons/year, Sulfur Oxides (SO_x) with 28,318.65 tons/year, Particulate Matter (PM) with 26,098.50 tons/year and Nitrogen Oxides (NO_x) with 13,074.69 tons/year.

Table 15. Total Emission in Davao City for CY 2022

Sources	TONS/YEAR				
	PM	CO	NO _x	SO _x	VOC
1. Stationary	8,672.79	4,691.00	3,752.78	27,770.53	48.12
2. Area	2,880.54	38.57	380.49	361.99	914.44
3. Mobile	14,545.17	114,613.64	8,941.42	187.12	55,253.64
Total Emission	26,098.50	119,343.21	13,074.69	28,319.65	56,216.20

Source: 2022 Davao City Emission Inventory, EMB XI

As shown in Figure 9, emissions from mobile sources contribute 58% to the total PM concentration, 89% to the total CO concentration, 65% to the total NO_x concentration, and 96% to the total VOC concentration in Davao City. For SO_x, the highest source of emission is stationary source which contributes 100% to the total pollutant concentration in the city.

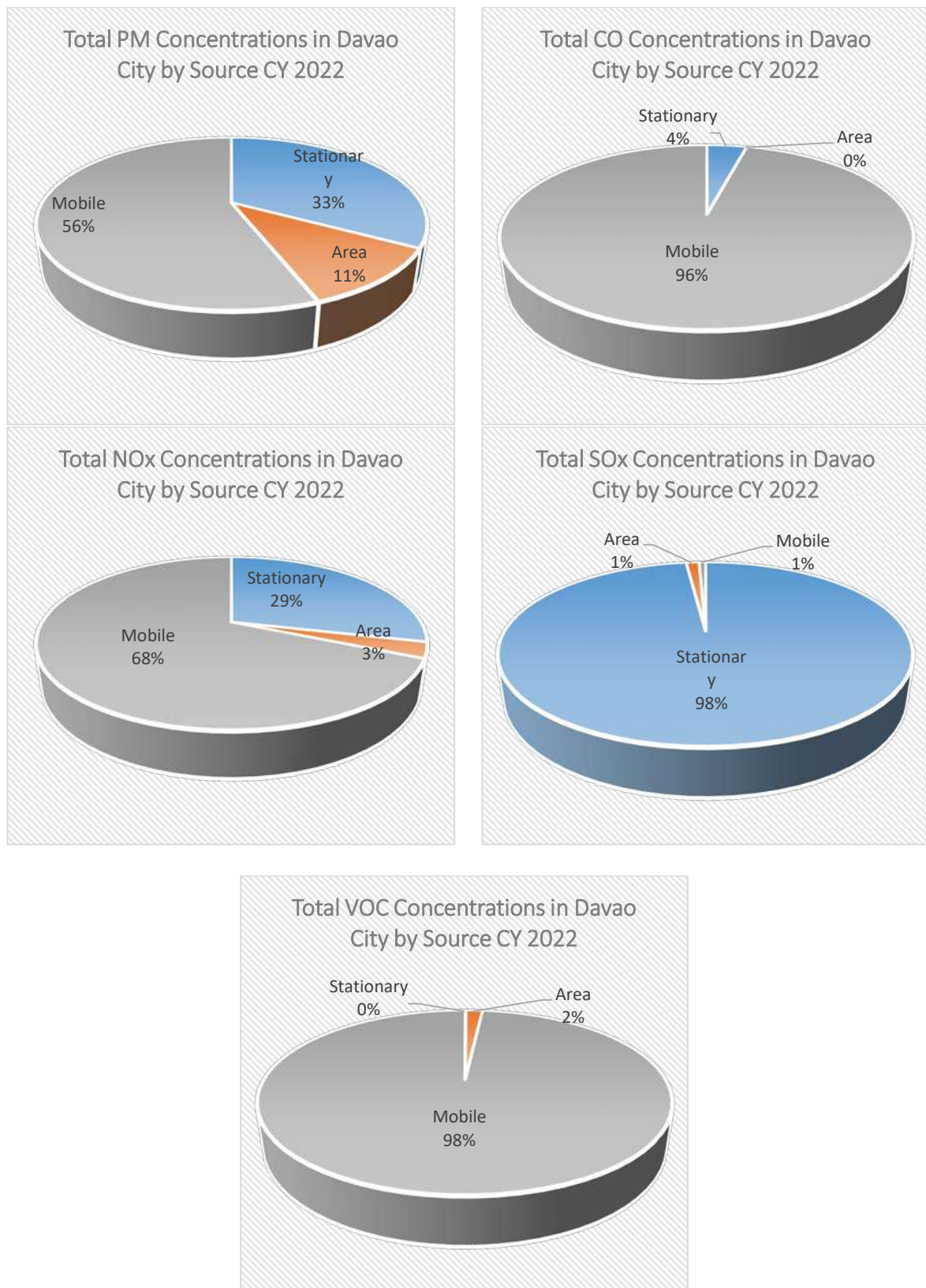


Figure 9. Total Emissions in Davao City per Pollutant per Source in Percentage

MANAGEMENT OF AIR POLLUTION SOURCES

Management of Stationary Sources

Stationary sources are being managed pursuant to the provisions of the Act and its Implementing Rules and Regulations, Memorandum Circular 2007-03, Department Administrative Order No. 2004-26, DENR Administrative Order No. 2007-22, and others.

Basic management is to require establishments that all of their air pollution sources must have an Air Pollution Control Device and have valid Permit to Operate Air Pollution Sources and Control Devices. The monitoring of stationary sources involves ocular inspections, source emission tests, and ambient source sampling tests. *(Existing sources are those installed prior to November 25, 2000)*

According to DAO 2000-81 or the Implementing Rules and Regulations (IRR) of RA 8749, installation of a Continuous Emission Monitoring System (CEMS) is mandatory for sources with the potential to emit quantities equal to or greater than 100 tons per year of any pollutants. To continuously measure the emission of pollutants from an affected facility or source, a CEMS is required. Large and environmentally significant sources also require dispersion modeling to demonstrate compliance with the source-specific ambient air quality standards prior to initial permitting.

The EMB Memorandum Circular 2007-03 provides that source emission testing shall take place twice a year for environmentally significant sources and large sources (boilers with a capacity equal to or greater than 251 hp), medium sources (boilers of 100 to 250 hp) must undergo emission testing once a year and small sources (boilers with a capacity of 99 hp and below) should undergo emissions testing every two years.

A total of 237 permits were issued for the year 2023 (new and renewal) by EMB XI.

Table 16. Permits Issued

	2023
Permits issued within Davao City Airshed (no.)	237

Management of Mobile Sources

In the management of mobile sources, all in-use motor vehicles and vehicles with updated/enhanced engines whose chassis are pre-registered with DOTC/LTO will only be allowed renewal of registration upon proof of compliance with the emission standard. The DOTC/LTO shall conduct the vehicle test utilizing the Motor Vehicle Inspection Station (MVIS) or its duly authorized and accredited Private Emission Testing Centers (PETCs) to ensure they pass the emission standards before their registrations are renewed. For check and balance, roadside inspections are being conducted by the enforcement unit of LTO and the Anti-Smoke Belching Unit (ASBU) of City ENRO.

PETC/PMVIC Monitoring

A total of 18 Private Emission Testing Centers (PETC) and Private Motor Vehicle Inspection Centers (PMVIC) were operating within the Davao City Airshed (LTO XI). EMB XI personnel conducted monitoring of PETCs from February to December 2022 to determine if the equipment being used is/are included in the EMB CO's List of Vehicle Testing Equipment and proper maintenance schedule is being done (such as calibration). As of December 2023, a total of 17 PETCs/PMVICs are still operational. The remaining 1 PETC was found to be permanently closed during the conduct of the site visit. Monitoring reports were also endorsed to Land Transportation Office XI for its information and reference.

Table 17. No. of PETCs Operating in Davao City

No. of PETC	18
PETC Monitoring conducted by EMB XI (no.)	17
Closed/Not operational (no.)	1



Figure 10. PETC Monitoring in Davao City

Anti-Smoke Belching Operation

One of the activities conducted by the Governing Board is the operation of Anti-Smoke Belching in coordination with the Anti-Smoke Belching Unit (ASBU) of the City Environment and Natural Resources Office – Davao City. This determines the compliance of drivers/operators to the vehicular emission standards. The operation covered from February to November 2023 every month. Citation tickets were issued to the operator/driver showing non-compliance with the Davao City Ordinance NO. 0280-06 otherwise known as Anti-Smoke Belching Ordinance. A total of 905 vehicles were tested. 379 passed and 526 failed during the entire period of testing.

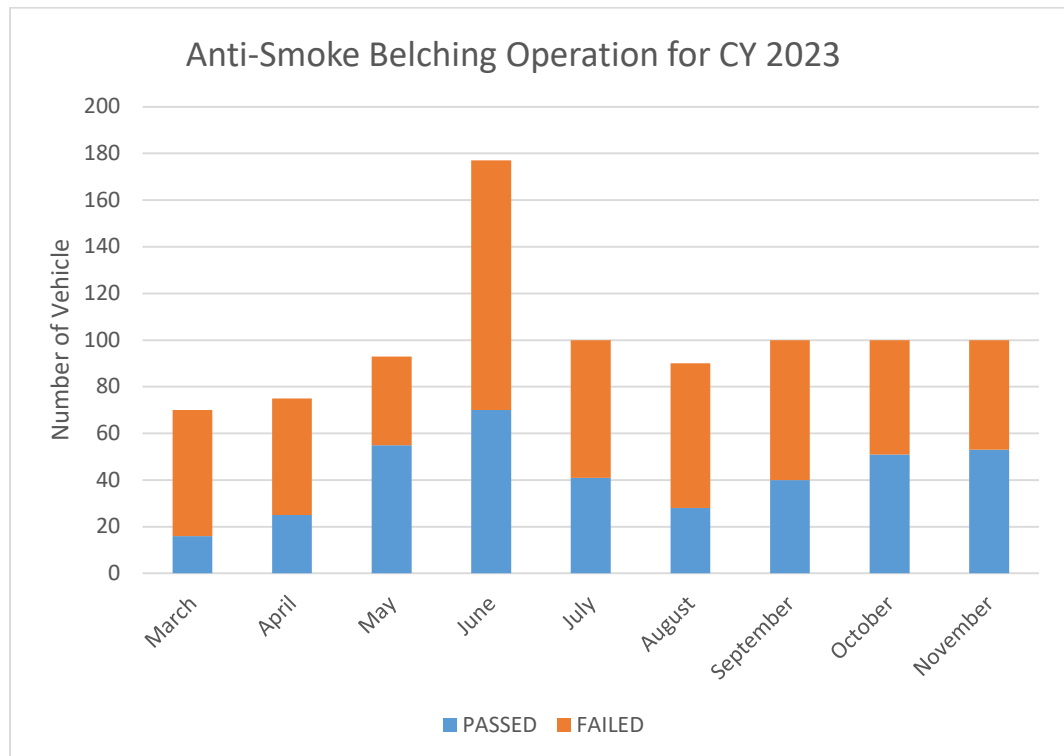


Figure 11. Monthly Roadside Inspection of Davao City Airshed Governing Board through City ENRO



Figure 12. March 2023 to November 2023 Anti-Smoke Belching Operation



Figure 13. Anti-Smoke Bleaching Operation in Celebration of CY 2023 Philippine Environment Month and Clean Air Month

Management of Area Sources

Open burning of garbage can cause serious health problems and damage to the environment. Pollutants that can be emitted through open burning include particulate matter, dioxin, heavy metals, and arsenic among others.

In the management of Area Sources, the City ENRO through Ordinance No. 0371-10 or Davao City Ecological Solid Waste Management Ordinance of 2009 prohibits the burning of solid wastes in accordance with the Solid Waste Management Program of Davao City.

OPERATIONALIZATION OF DAVAO CITY AIRSHED

The whole area of Davao City was initially designated as an Airshed through Memorandum Circular no. 2003-02 dated February 12, 2003. Currently, the Davao City Airshed is composed of thirty-one (31) members coming from National Government Agencies (NGAs), Local Government Units (LGUs), Non-Government Organizations (NGOs), Industries, and Private Sectors including Academe, Homeowners/Neighborhood Association, Network Transport Organization and Drivers Association. In November 2013, a 5-Year Action Plan (2014-2019) was approved by the Governing Board which focuses on the three (3) Key Result Areas: 1) Research, evaluation, monitoring & formulation of policies, 2) Advocacy to public awareness and elicit commitment and participation, and 3) Resource generation and management to support activities within the airshed.

In 2023, the Governing Board has implemented targets under the 5-Year Action Plan (2020-2024) regarding the activities intended to be undertaken concerning airshed operationalization. Activities include a series of meetings to come up with strategies for maintaining the air quality in the airshed; strengthening the monitoring of vehicular emissions; and continuous Information, Education, and Communication (IEC) campaign.

In the same year, eight (8) board resolutions were approved by the Governing Board. The resolution was carried out to further strengthen the monitoring of Air Quality in Davao City (see annex A).

Technical Working Group	
Meeting Regarding Discussion of Details of the Memorandum of Agreement (MOA) for the Adopt a Continuous Ambient Air Quality Monitoring Station Program	January 19, 2023
Engine Servicing and Motor Vehicle Emission Control Technician (MVECT) Training.	February – March, 2023
2-Day Workshop on Air Pollutant Mapping in the Davao City Airshed	March 23-24, 2023
Training/Workshop on Ambient Air and Stack Sampling, and Airshed Designation and Operationalization vis-à-vis Environmental Law.	March 28-30, 2023
Public Hearing for the Designation of Davao City Airshed as Attainment Area for Particulate Matter 10	January 19, 2023

Governing Board	
Davao City Airshed Governing Board Meeting at Ritz Hotel at Garden Oases, Bo. Obrero, Davao City	June 16, 2023
Davao City Airshed Governing Board Meeting at Waterfront Insular Hotel, Lanang, Davao City	November 24, 2023

Table 18. Meetings conducted (Governing Board, Executive Committee, and Technical Working Group)



Figure 14. 1st Semester Davao City Airshed Governing Board Meeting



Figure 15. 2nd Semester Davao City Airshed Governing Board Meeting



Figure 16. Meeting for the Discussion of Details of MOA for Adopt an Ambient Air Quality Monitoring Stations Program



Figure 17. Engine Servicing and MVECT Training



Figure 18. 2-Day Workshop on Air Pollutant Mapping in the Davao City Airshed



Figure 19. Training/Workshop on Ambient Air, Stack Sampling, and Airshed Designation

Figure 20. Public Hearing for the Designation of Davao City Airshed as Attainment Area for PM₁₀

Table 19. 2023 Board Resolutions Approved

Board Resolutions Approved	Subject
Board Resolution No. 1 series of 2023	Confirmation of the Representatives of Active Sectoral Groups through Securities and Exchange Commission (SEC) Certification and Appointment of Permanent and Alternate Representatives for the Davao City Airshed Governing Board
Board Resolution No. 2 series of 2023	To Include the Davao Inventors Association (DIA) as a Regular Member of the Davao City Airshed Governing Board, Represented by Mr. Virgilio M. Sangutan
Board Resolution No. 3 series of 2023	Proposal that Non-Active Members be Removed from the Davao City Airshed Governing Board List of Members
Board Resolution No. 4 series of 2023	Creation of Technical Working Group for the Davao City Airshed 5-Year Action Plan CY 2025 to CY 2029
Board Resolution No. 5 series of 2023	Special Meeting of the Technical Working Group to Draft the Davao City Airshed 5-Year Action Plan CY 2025 to CY 2029
Board Resolution No. 6 series of 2023	The Inclusion of Women's Groups or Environmental Groups with GAD-Related Programs in the Governing Board be Deferred
Board Resolution No. 7 series of 2023	Requiring the Sectoral Members to Submit General Information Sheet (GIS) Duly Received by Securities and Exchange Commission (SEC)
Board Resolution No. 8 series of 2023	Terms of Membership of Active will be Renewed while Membership of inactive Sectoral Members will be Subject to Evaluation and Communication for their Interest to Sit as Governing Board Members

CHALLENGES

- 1) For the management of mobile sources, the Anti-Smoke Belching Unit of City ENRO Davao City only has one (1) opacimeter which enables them to monitor emissions from vehicles traversing in one specific location per operation. There were no gasoline-powered vehicles subjected to testing of hydrocarbons and carbon monoxide since CY 2021 up to the present because the gas analyzer (FGA 1400) was under repair.
- 2) Several areas in the Davao City Airshed are exposed to vehicular emissions for more than eight (8) hours, however, it is beyond the spatial scale of representativeness of the current air quality monitoring stations.
- 3) The Davao City Airshed's automatic stations have been operational for more than ten years, thus some components are already malfunctioning, such as the data logger system due to system errors, requiring an additional budget for its upgrades. Requests for budgets for the station's maintenance are challenging since there is a limited budget available.
- 4) One of the data gaps in the Emission Inventor was the different proforma and coverage as what is required in the EI.

RECOMMENDATIONS

Executive and Legislative Action

To further improve the air quality of the region and its airshed, the following actions are hereby recommended:

1. Observe day-to-day monitoring, which includes routine maintenance checks of the equipment at automatic stations.
2. To have valid trending data on the Davao City Air Quality Monitoring Network, appropriate sampling methods and frequency of monitoring must be followed, as specified by the updated procedural manual on data handling protocol for criteria air pollutants derived from EMB Memorandum Circular 2021-07.
3. A copy of this report should be forwarded to LGU-Davao City and presented to the Davao City Airshed Governing Board to recommend necessary actions to be undertaken to improve air quality in the city.

Regulatory Actions of Concerned Government Agencies

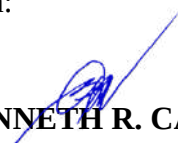
1. Updates on the transition from private emission testing centers (PETC) to private motor vehicle inspection centers (PMVIC) from the Land Transportation Office (LTO) and the Department of Trade and Industry (DTI).
2. In accordance with Clean Air Act requirements, the Department of Energy should strengthen its monitoring of fuel products.
3. To monitor gasoline-powered vehicles for hydrocarbon and carbon monoxide emissions, a gas analyzer needs to be requested from the LGU Davao City or the Land Transportation Office (LTO).
4. As part of the data-gathering process, the concerned agencies must respond promptly to the request for emission inventory sources. It is to utilize the data requested from them in the preparation of reports. Moreover, EMB XI will be providing a proforma that is useful for EI; hoping it will be adopted.

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